v12: Budget-Adaptive Self-Configuration with Validation-Gated Selection for Unseen Data

Sebastian Seifert, Luis Melzer & Leander von Gottberg

1 Motivation

Our proposed pipeline (v10/v11.1) measures the unseen data first, gates every decision against a robust incumbent default, and spends all remaining time on training. A three-way comparison on the public datasets has since confirmed the design premise: the informed-defaults floor finishes within 0.6 points of the incumbent, while a faithful reproduction of the 2025 winner [1] loses 69.9 points net, regularised evolution loses 25.9, and a per-stage hill-climb adopted a wrong architecture on every blind hold-out it was offered. Independent work published on the same benchmark reaches the same wall from the other side: learned cross-dataset performance predictors degrade to near-zero rank correlation exactly on the misaligned datasets (Kendall τ 0.19 and 0.10) [3, 4]. Architecture *search* is not where the remaining accuracy is.

To locate where it is, we ran a 93-run controlled screening campaign on nine datasets spanning every input archetype (boards, reasoning grids, text encodings, satellite and natural images, cellular automata, scientific tensors): candidate operations from a brain-inspired architecture, structural re-parameterisation, depth ladders in two families, weight averaging, seed ensembling, test-time augmentation, and five training-recipe toggles, each against an identical fixed-arch default at matched budget. Three findings define v12. **(i)** The wins are in the recipe and at inference time, not in the architecture: parameter-free stochastic augmentation in the style of TrivialAugment [6] gains +2.3 points on average (8/9 datasets non-negative, +13.7 on the cellular automaton), and averaging the logits of K=3 seeds gains +1.5 (8/9); stacked, they lift one dataset 4.3 points above the incumbent. Every architectural intervention washed out or regressed, and depth closes none of the one known family gap. **(ii)** Several techniques have dataset-dependent sign: flip-TTA gains +2 on flip-invariant grids and loses 12 on text encodings; sharpness-aware minimisation wins small on four datasets and collapses one by 19 points; weight averaging of independently trained seeds can destroy a model outright. Such techniques are deployable only behind a per-dataset decision, and our earlier results show that decision must be *measured*, never predicted [5]. **(iii)** Budgets and hardware are unknown at runtime (the final phase pools 24 hours across all datasets on an unspecified device), and wall-clock is the wrong unit for stage planning: the same 4-hour budget that helps a slow dataset silently *mis-sizes* the cosine schedule on a fast one, because a fixed epoch cap binds long before the clock and the anneal never completes.

2 Research Questions

RQ1. Do recipe and inference-time gains measured on a controlled testbed transfer additively onto the routed, full-strength pipeline, or do they shrink on strong backbones the way short-training mirages do? **RQ2.** Can a runtime throughput calibration (converting the remaining clock into *effective trainings* on the actual device and data) replace wall-clock rungs, so one pipeline scales from a 30-minute smoke test to an 8-hour final run and across GPUs of very different speed? **RQ3.** Can techniques with dataset-dependent sign (ensembling, test-time augmentation, aggressive regularisation) be made safe-by-construction on unseen data through a final validation-gated selection stage, at what budget does each begin to pay, and how much of the oracle gain does the gate capture?

3 Approach

v12 keeps the measured-and-gated core of v11.1 and adds three mechanisms, each validated by the screen (Fig. 1). The search principle is unchanged: every choice competes against an incumbent default and is adopted only on a measured win. (0) *Calibrate*: a few training steps of the default network on the real data measure the per-step time on the actual device; the remaining clock divides into E, the number of full trainings that fit. E, not hours, selects which stages run, and it is re-projected from measured times after every finished stage, so a slower-than-expected dataset demotes itself mid-run instead of overrunning. A fast device reaches the full program at budgets where a slow one runs only the core; no hardware detection is involved. (1) *Measure and route*: the training-free fingerprint and family routing of v10, unchanged. (2) *Recipe*: stochastic single-operation augmentation (tensor-safe operations, one per batch, no policy search [6]) becomes part of the default recipe, and the cosine schedule is sized to the *realisable* epoch count $\min(projectedepochs, cap)$ so the anneal always completes, fixing the mis-sizing found at long budgets. (3) *Probes* (moderate E): the aug-safety and verified attention probes of v11.1. (4) *Seeds and specialists* (large E): the surplus budget trains K seeds of the chosen configuration; gated specialist arms may join (a linear-attention sequence model on short encoded boards behind its runtime viability check; candidate operations only through verify-before-adopt). (5) *Selection*: at the end, the pipeline enumerates inference configurations (best single seed, logit ensemble, each with and without flip-TTA), scores each on held-out validation, and ships the winner. The single best seed is the built-in null hypothesis, so a technique that harms this dataset is filtered by measurement rather than by a prediction; the one augmentation failure observed in the screen is validation-visible.

4 Required Steps

(1) Controlled screening campaign across operations, architectures, recipes, and budgets (done). (2) Integration test: re-measure the adopted deltas on the routed pipeline against v10/v11.1 on all public datasets (RQ1). (3) Rung-calibration matrix: {core, +probes, +seeded selection} × {0.5, 2, 6 h} on three datasets to place the E thresholds

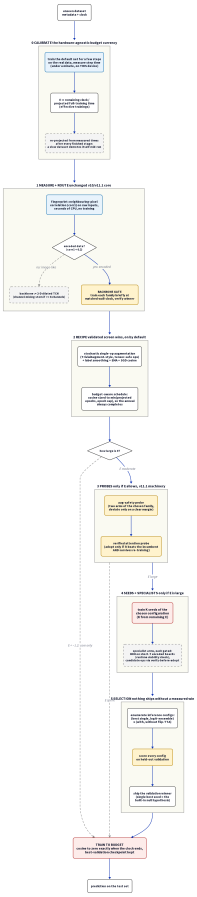


Figure 1: The v12 decision pipeline, run once per unseen dataset. Diamonds are runtime decisions, dashed paths are the abstain-to-default routes. New against v11.1: the throughput calibration (stage 0) that converts the clock into effective trainings E and re-projects after every stage, the validated default recipe (stage 2), and the validation-gated selection stage (stage 5) that turns dataset-dependent techniques into measured, safe choices.

empirically (RQ2). (4) Ablations: flip-free augmentation variant, per-sample versus per-batch application. (5) Full multi-seed evaluation. (6) Packaging and stress tests from 5 minutes to 24 hours under the mock grading harness.

5 Evaluation Strategy

Primary evaluation uses the 13 public competition datasets plus three self-built competition-style datasets, with the official adjusted score and at least three seeds for final configurations. Baselines: v10, the v11.1 floor, and an oracle that always picks the best inference configuration in hindsight, bounding the achievable selection gain (RQ3). Every stage writes its measurements, margins, and choices to a per-dataset decision record, so gains and losses remain attributable to individual decisions. Robustness is evaluated across budgets (30 minutes to 12 hours), across devices of different throughput, and against published competition bests [2].

6 Potential Extensions

Under the same validation gate, the seed pool can diversify: optimiser and regularisation variants (AdamW, sharpness-aware minimisation) as additional arms where E permits, since the screen shows they help a minority of datasets and the gate filters the rest. Beyond logit averaging, parameter-space composition of family specialists remains a cheap candidate for the largest budgets.

References

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